

MANIPAL UNIVERSITY JAIPUR
Department Computer Science and Engineering
Course Hand-out

A. Basic Details:

Programme Name:	B.Tech-Computer Science and Engineering
Course Name:	Problem Solving Using Computers
Course Code:	CSC1101
LTPC (<i>Lecture Tutorial Practical Credits</i>):	2 1 0 3
Session:	JULY 2025-NOVEMBER 2025
Class:	B. Tech 1 st Year
Course Coordinator:	Dr.Jeyakrishnan.V
Course Instructor(s):	Dr Manoj Bohra Dr. Rajat Goel Dr. Ashish Sharma Dr. Siddhanta Kumar Singh Dr Krati Dubey Mr. Anurag Bhatnagar Dr Suresh Kumar Jha Dr. Bhawna Sharma Dr. Virendra Kumar Meghwal Mr. Arvind Kumar Mehta Mr.Jay Shankar Sharma Dr. Shweta Gangrade Dr. Vikas kumar boradak Dr Arun Johar Dr. Basudha Dewan Ms. Tripti Dr. Atul Kumar Verma Ms. Vipasha Ms. Harshika Dr. Princy Randhawa
Additional Practitioner(s) – if any (<i>Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.</i>):	To be identified and appointed later/ or at the beginning of the semester

B. Introduction:

Problem solving using computers course focuses on basic computer fundamentals, number system and programming in C fundamentals. By means of C language students learn to write a set of instructions to create a program so that desire output can be generated by computer.

C. Course Outcomes: 4-6 course outcomes as per guidelines and characteristics specified in the previous section. The general template of writing course outcomes is mentioned as follows:

<i>CO Statement</i>	<i>CO</i>	<i>Level</i>	<i>Target Attainment %</i>	<i>Target Attainment level</i>
Explain the fundamental concepts of integer and floating-point representations and core elements of the C programming language.	CO1	2		
Apply C programming constructs such as data types, variables, expressions, control flow, loops, and functions to write structured programs.	CO2	3		
Demonstrate 1-D and 2-D arrays, strings, and structures for various programming applications. .	CO3	3,4		
Solve problems using functions, recursion, file handling, and basic pointer operations.	CO4	3		
Analyze and debug C programs to identify logical errors and improve code efficiency and modularity.	CO5	5,6		

Here, the level indicates the cognitive level of Revised Bloom's Taxonomy.

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes: Mention program outcomes and program specific outcomes of the program.

[PO.1]. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

[PO.2]. Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4]. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5]. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

[PO.6]. The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues, and the consequent responsibilities relevant to professional engineering practice.

[PO.7]. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practices.

[PO.9]. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of engineering. and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12]. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Quizzes (Best of Two: 2x10=20) and Assignments (Two: 2x5=10)	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

Attendance (Formative)	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The allowance of 25% includes all types of leaves including medical. leaves.
Make up Assignments (Formative)	Students who miss a class will have to report to the teacher about the absence. A makeup assignment on the topic taught on the day of absence will be given which has to be submitted within a week from the date of absence. No extensions will be given on this. The attendance for that day of absence will be marked blank, so that the student is not accounted for absence. These assignments are limited to a maximum of 5 throughout the entire semester.
Homework/ Home Assignment/ Activity Assignment (Formative)	There are situations where a student may have to work in home, especially before a flipped classroom. Although these works are not graded with marks. However, a student is expected to participate and perform these assignments with full zeal since the activity/ flipped classroom participation by a student will be assessed and marks will be awarded.

F. Syllabus: The syllabus of the course as specified in the program curriculum need to be included here. Please do not modify the contents of the syllabus unless approved by BoS. In addition, reference and textbooks need to be mentioned for ready reference guide to the learners.

Module	CONTENTS
1	Introduction to assembly and high-level programming languages (programming paradigms), Number System, Basic concepts of integer and floating-point number representations;
2	Fundamentals of Data Types and Variables in Programming: Data types, constants, variables, Enumerated datatypes, Storage Class
3	Understanding Operators, Expressions, and Control Statements: Operators, expressions, Control Statements and branch statements, iteration statements
4	Working with Arrays -1D, 2D and Pointers in Programming: Arrays and pointers
5	Functions and parameter passing, Recursion, strings and Structures in Programming: Structure – Nested structures, Array of structures
6	File Handling, Standard Library Functions, and Elementary Data Structures: File Handling, some standard library functions

References:

- E. Balagurusamy, “Programming in ANSI C”, 7th Edition, McGraw Hill Publication, 2017.
- Y. P. Kanetkar, “Let us C”, 16th Edition, BPB Publication, 2017.
- B. W. Kernighan, D. M. Ritchie, “The C Programming Language”, 2nd Edition, Prentice Hall of India, 2014.
- B. Gottfried, “Schaums Outline Series: Programming with C”, 3rd Edition, McGraw Hill Publication, 2012.

G. Lecture Plan: The lecture plan of the course indicates clear schedule of content delivery, session outcome, corresponding CO, mode of delivery, and mode of assessment. The general template of lecture plan is mentioned as follows:

Lecture No.	Topic	Session Outcome	Corresponding CO	Mode of delivery	Mode of assessing CO
1	Introduction and Course Hand-out briefing.	To acquaint and clear teachers' expectations	NA	NA	NA
1	Von Neumann architecture, programs, assembly language, high level programming languages	To acquaint basic knowledge of computer system.	1101.1	Lecture based teaching learning	
2,3,4	Number System: Binary, Octal, Decimal, Hexadecimal and their conversions.	To learn process to solve any complex problem	1101.1	Lecture based teaching learning/ Individual learning/ self-study	MidTerm, Quiz & End Term
5,6	Data types (table including range, memory, and format specifier), Operators: arithmetic, relational, logical, assignment.	Implementation of various data type	1101.1	Lecture based teaching learning	MidTerm, Quiz & End Term
7,8	Storage Class, Enumerated datatype	To gain knowledge of storage class and enumerated data types	1101.1	Technology based learning	MidTerm, Quiz & End Term
9,10	Bitwise, conditional	Implementation of various conditional, bitwise operations	1101.1	Technology based learning	MidTerm, Quiz & End Term
11	Type-cast, size of, comma	Implementation of various arithmetic operations	1101.1	Lecture based teaching learning	MidTerm, Quiz & End Term
12	Operator precedence and associativity, type conversion	Implementation of precedence in programming	1101.1	Learning through problem-solving	MidTerm, Quiz &

					End Term
13,14	Input and output statements (formatted and unformatted): printf, scanf, gets, puts, getchar, putchar	Implementation of input and output statements	1101.2	Lecture based teaching learning	MidTerm, Quiz & End Term
15,16	Decision statements: if, if-else, nested if-else, if-else ladder	Implementation of decision statements	1101.3	Learning through problem-solving	MidTerm, Quiz & End Term
17	Switch, break statement	Learning the implementation of switch and break	1101.3	Learning through problem-solving	MidTerm, Quiz & End Term
18	Repetitive structures: for, while loops	Learning the implementation of looping	1101.3	Lecture based teaching learning	MidTerm, Quiz & End Term
19,20	Repetitive structures: do-while loops, nested loops	Learning the implementation of looping	1101.3	Learning through problem-solving	MidTerm, Quiz & End Term
21	More examples on Nested loops of various types	Learning the implementation of looping	1101.3	Lecture based teaching learning	MidTerm, Quiz & End Term
22	Continue and break statements	Describe the usage of continue and break	1101.3	Lecture based teaching learning	MidTerm, Quiz & End Term
23	1-D array: definition, declaration, initialization, input array, output array	Describe and define array of various data type	1101.3	Lecture based teaching learning	MidTerm, Quiz & End Term

24,25	1-D array: Examples of array operations	Describe and define array of various data type	1101.3	Technology based learning	MidTerm, Quiz & End Term
26	1-D character array: character array, string, standard functions for string manipulation	Describe and define array of various data type	1101.3	Lecture based teaching learning	MidTerm, Quiz & End Term
27	1-D character array: user defined functions of standard string manipulation functions	Describe and define array of various data type	1101.3	Lecture based teaching learning	MidTerm, Quiz & End Term
28	Application of 1-D array: Linear Search and Bubble Sort	Describe use of linear array	1101.3	Technology based learning	MidTerm, Quiz & End Term
29	2-D array: definition, declaration, initialization, input array, output array, one simple program	Describe and define array of various data type	1101.3	Technology based learning	MidTerm, Quiz & End Term
30	2-D array: More examples on 2-D arrays	Describe and define array of various data type	1101.3	Peer teaching	MidTerm, Quiz & End Term
31	2-D array: More examples on 2-D array	Describe and define array of various data type	1101.3	Technology based learning	MidTerm, Quiz & End Term
Mid Term Examination					
32	Pointers: Introduction	Describe functionality of pointers in programming	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
33	1-D Array and pointer	Implementation of 1-D array with pointer	1101.4	Peer teaching	MidTerm, Quiz

					& End Term
34	More examples on pointer and operations using pointers	To acquaint more knowledge about pointers	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
35	Some more examples on pointer and operations using pointers	To acquaint more knowledge about pointers	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
36	Functions: introduction to functions, function prototype, call, definition	Describe importance of function and modular programming	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
37,38	Call by value and Call by reference	Describe importance of function and modular programming	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
39	More examples using functions	Describe importance of function and modular programming	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
40	Examples using functions continued	Describe importance of function and modular programming	1101.4	Lecture based teaching learning	MidTerm, Quiz & End Term
41	Structures: definition, declaration, initialization	Describe usage of Structure	1101.3	Technology based learning	MidTerm, Quiz & End Term
42	Structures: array of structures	Describe usage of Structure	1101.3	Technology based learning	MidTerm, Quiz & End Term

43	Union, difference between union and structures	Describe usage of union	1101.3	Technology based learning	MidTerm, Quiz & End Term
44	File handling: introduction, operations on files, opening modes	Describe usage of file handling with various operations and modes	1101.5	Peer teaching	MidTerm, Quiz & End Term
45	File handing function	Describe usage of file handling with various operations and modes	1101.5	Peer teaching	MidTerm, Quiz & End Term
End Term Examination					

Outside Class Engagement Activities:

Type of Activity	Difficulty Level	Hours Engaged
Assignment 1	Medium	2-3 hours
Quiz 1	Medium	30min-1 hour
Assignment 2	Medium	2-3 hours
Quiz 2	Medium	30min-1 hour
Quiz 3	Medium	30min-1 hour
Midterm Examination	Medium	1 hour 30 min
End term Examination	Medium	3 hours
Hackathon/Industry Visit Other Events Participation	-	1-5 days

Important Update:

1. As per NAAC Reforms 2024- Binary Accreditation Stakeholder Consultation (University), the mode of delivery should include the followings:

1. Lecture based teaching-learning
2. Group- teaching and learning
3. Individual learning/ self-study
4. Inquiry based learning
5. Kinaesthetic learning
6. Game Based learning
7. Expeditionary learning
8. Technology based learning
9. Peer teaching
10. Learning through problem-solving

11. Flipped Classroom, etc.

It is expected that, MUJ-TEC will conduct workshops on these pedagogical in the coming time for more clarity.

2. In addition, some lectures should be marked as * that indicates the lectures to be covered by additional practitioners (Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.)

H. Course Articulation Matrix: Course articulation indicates the mapping of CO statement(s) to some PO/ PSO statement(s). Here value 0 indicates no correlation, 1 as low correlation, 2 as moderate correlation, and 3 as substantial / strong correlation.

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES												CORRELATION WITH PROGRAM SPECIFIC OUTCOMES		
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CSC 1101.1	Understand algorithm development, elements of programming languages and designing flow charts.	2	1										1	2	1	1
CSC 1101.2	Apply the basic programming concepts such as tokens, data types, operators and control statements for implementing programs.	2		1									2	1		
CSC 1101.3	Describe and analyze the concepts of array data type (1D and 2D), functions, structure and union.	3	1										2	1		
CSC 1101.4	Illustrate the concept of pointers and file Handling	2		2									1			
CSC 1101.1.5	Creating algorithms or pseudo code to solve real life problems using programming constructs	3	2	1									3	3	1	1